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Structural Causal Models

A Shared Language for Scientific Inquiry



Julian **Frattini**, Ph.D.

Chalmers University of Technology, Sweden

<https://julianfrattini.github.io/>



JulianFrattini/**bda4sci**

Public

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Directed Acyclic Graphs

A Syntax to Formalize Causal Assumptions

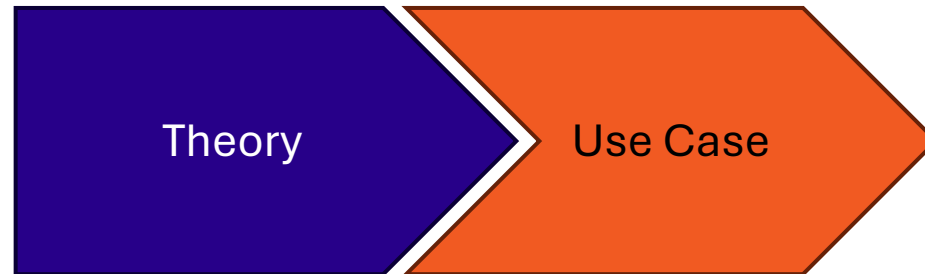
Causal Modelling

A (structural) causal model is a **conceptual model** that *represents the causal mechanisms of a system* [1]. They serve two main purposes [2]:

1. **Visualizing causal assumptions** and determining testable implications.
2. **Proving the possibility** of computing causal effects from observational data.

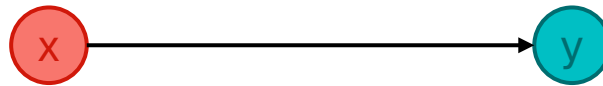
Structure of this Seminar

Every lesson consists of its theory and one or more use cases.



Directed Acyclic Graphs

Causal models can be realized as directed, acyclic graphs (DAGs).



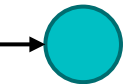
exposure



response



(assumed) causal
relationship



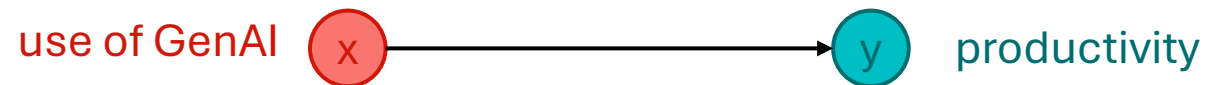
phenomenon

Formulating Casual Hypotheses

DAGs can visualize a causal hypothesis graphically

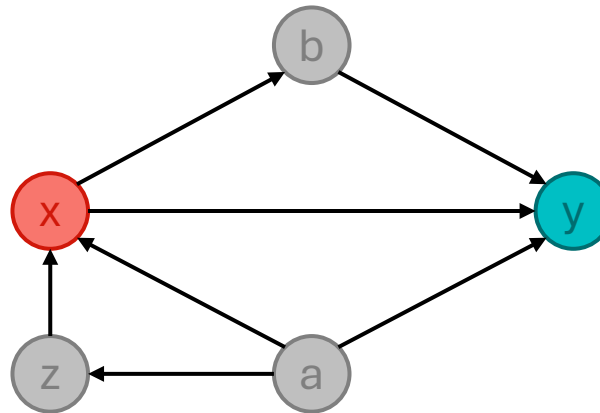
”

RQ: Does the use of **generative AI (GenAI)** have an **effect** on the **productivity** of developers?



Directed Acyclic Graphs

Causal models can be realized as directed, acyclic graphs (DAGs).



exposure



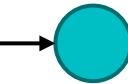
response



other factors



(assumed) causal
relationship



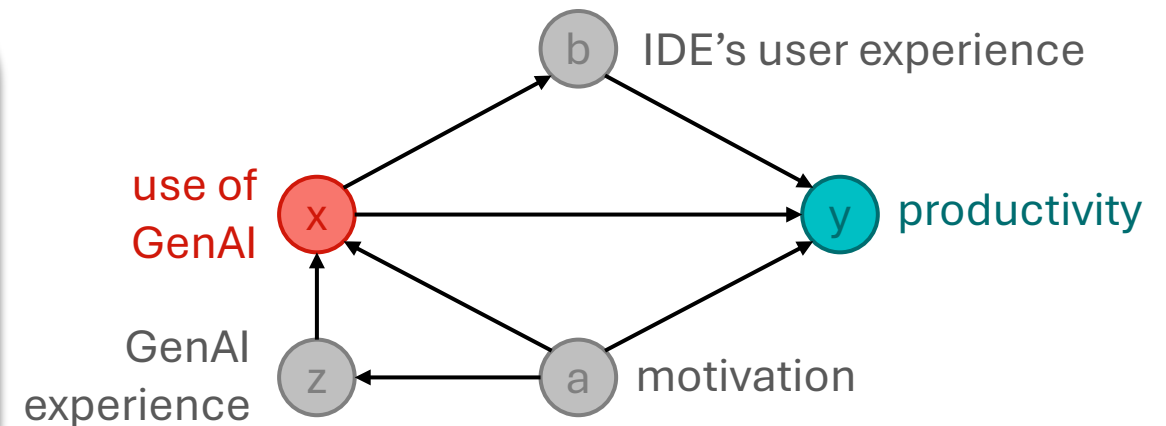
phenomenon

Formulating Complex Hypotheses

DAGs convey complex relationships between multiple variables in a simple and straightforward manner.

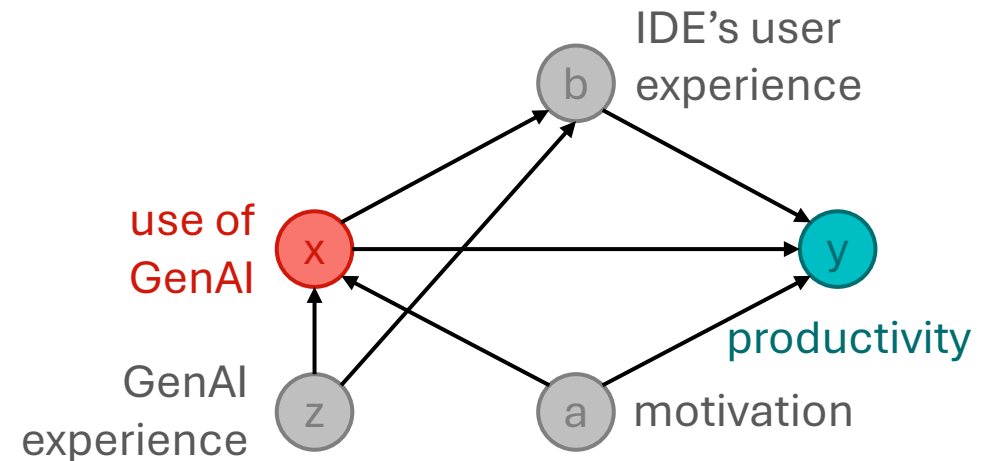
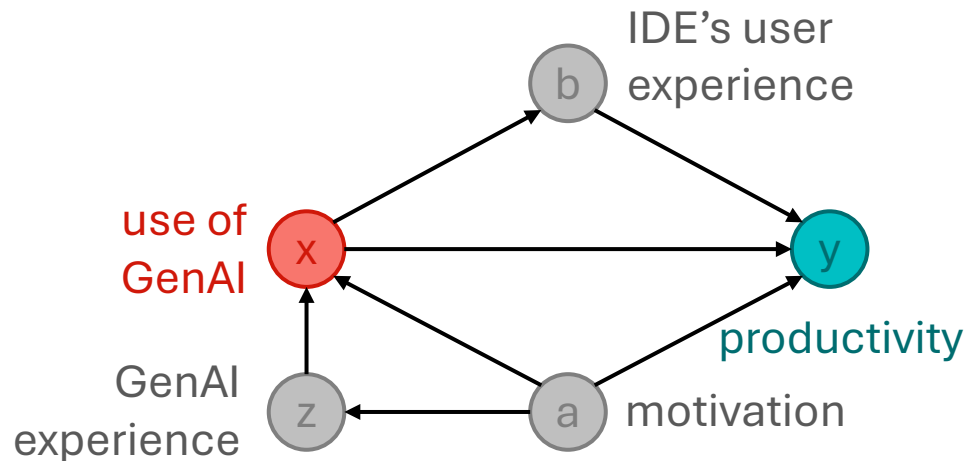
”

RQ: Does the use of **generative AI (GenAI)** have **an effect** on the **productivity** of developers, assuming that **motivation** affects both usage (moderated by **GenAI experience**) and productivity, and accounting for an indirect effect of the **IDE's user experience**?



Disputing Causal Hypotheses

The explicit formulation of causal hypotheses in DAGs makes it easy to compare and contrast assumptions.

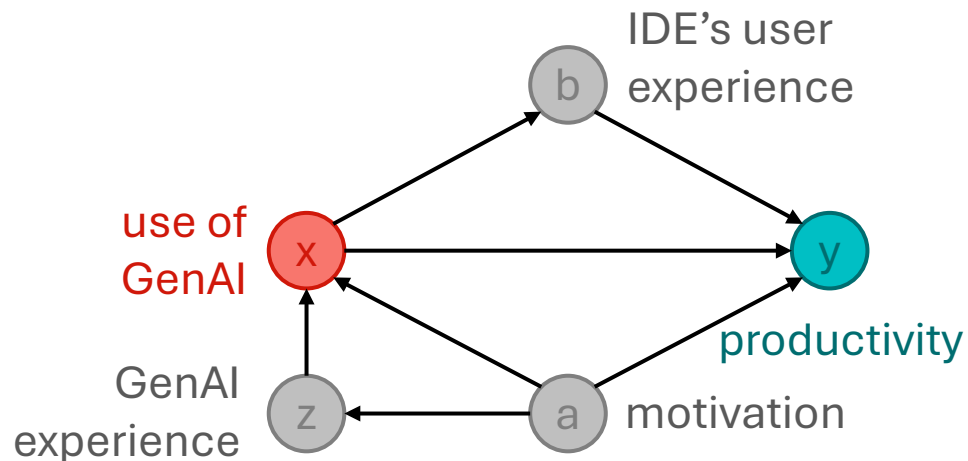


These comparisons can inform the design of an empirical study to verify, which of the two hypotheses reflects reality better.

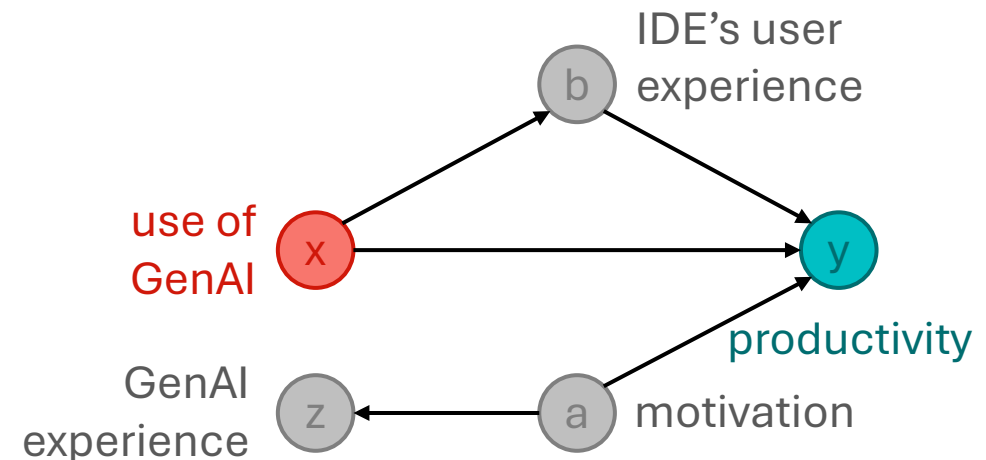
Distinguishing Study Types

The causal DAG also reveals controlled vs. observational studies: a controlled study is indicated by a DAG where no variable affects the exposure.

Observational study



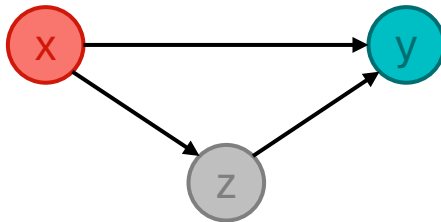
Controlled Experiment



Types of Relationships

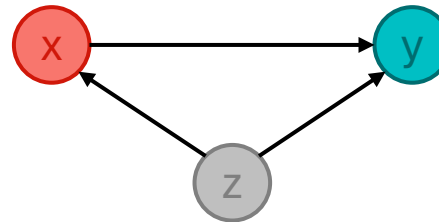
When a DAG involves more than two variables, certain relationships can emerge:

Mediation



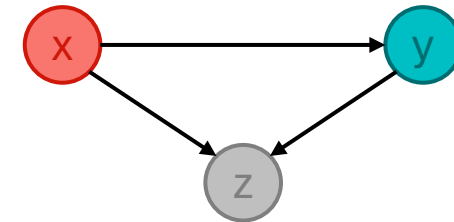
z *mediates* the effect of x on y . There is a direct ($x \rightarrow y$) and an indirect ($x \rightarrow z \rightarrow y$) effect.

Confounding



z *confounds* the effect of x on y . An observed effect ($x \rightarrow y$) might be caused by the confounder ($x \leftarrow z \rightarrow y$).

Collider

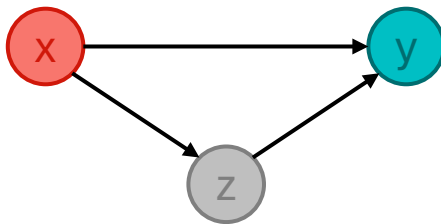


z is a common effect of x and y .

Dealing with Variables

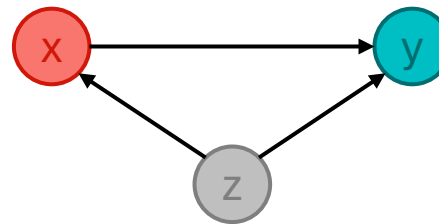
Based on the type of relationship between variables, these additional factors need to be treated differently.

Mediation



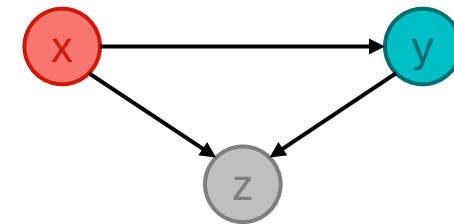
You **can** control z if you want to distinguish the direct from the indirect effect

Confounding



You **must** control z to avoid confounding

Collider

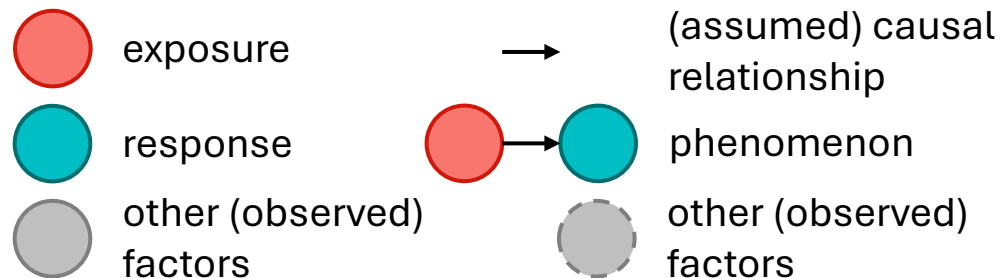
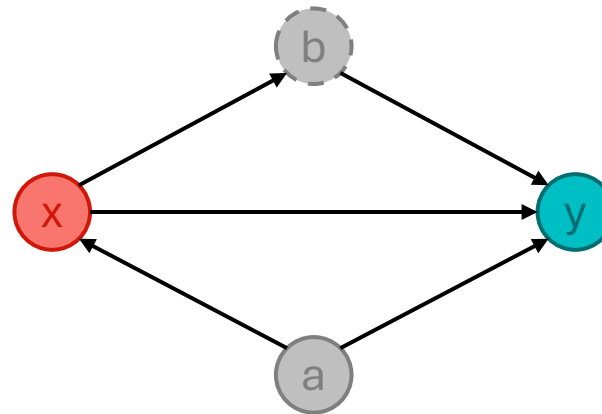


You **must not** control z to avoid confounding

Control can either be by design (fixing variables) or by analysis (stratifying the data).

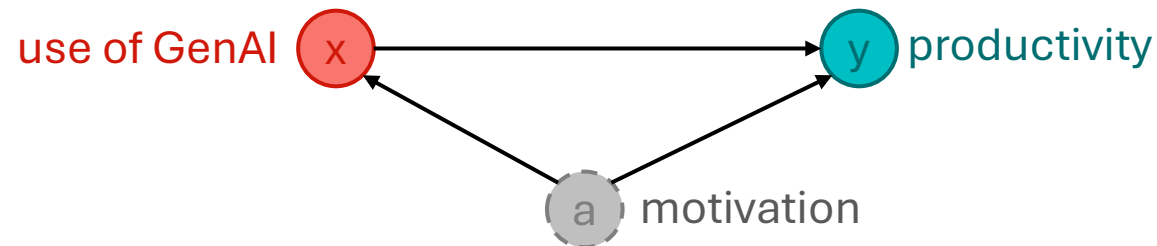
Encoding Observability

Nodes in DAGs can indicate whether a variable is observable or not.



Determining Study-ability

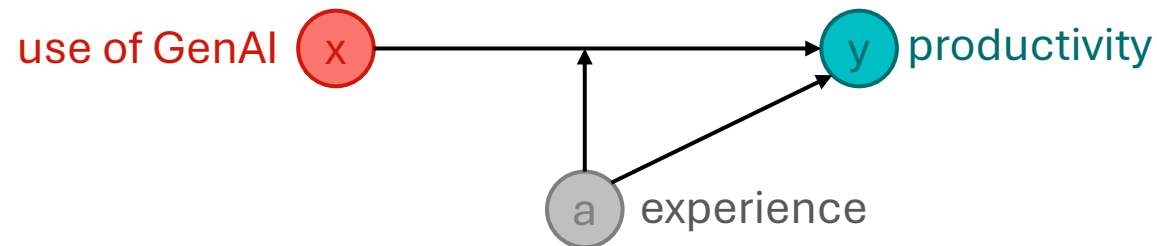
Causal DAGs allow deciding whether it is even possible to determine the (unbiased) causal effect of a phenomenon.



If a phenomenon has an unobserved confounder, then the causal effect of the exposure on the outcome cannot be estimated from observational data.

Visualizing Moderation

Factors can not only affect other factors, but also other relationships in a process called moderation.

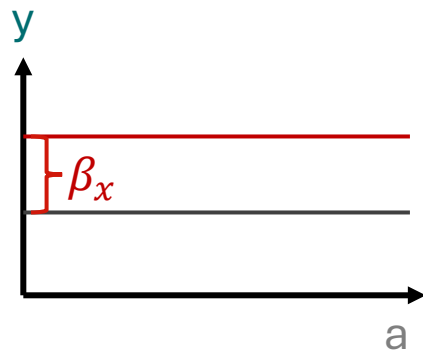
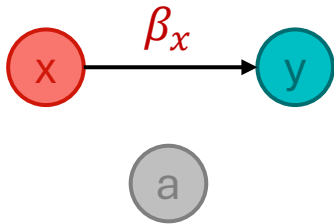


A moderator changes the effect that one variable has on another.

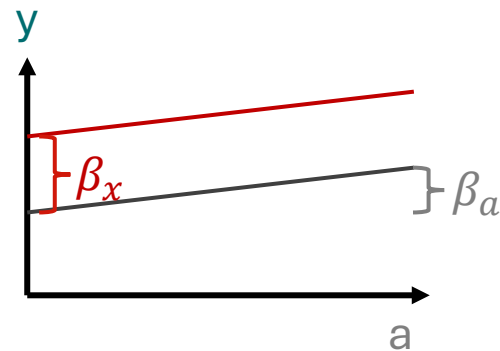
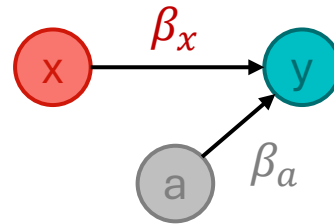
Determining Moderation

The existence of a moderator evokes a distinct effect pattern.

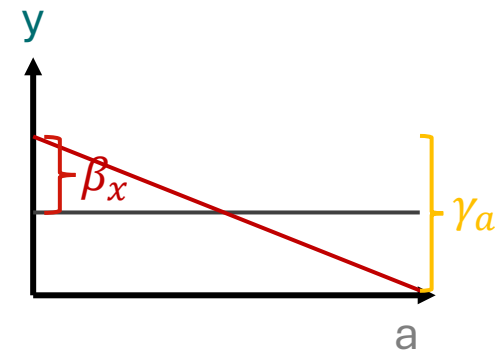
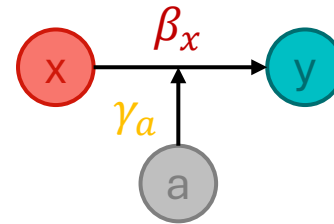
no covariation,
no moderation



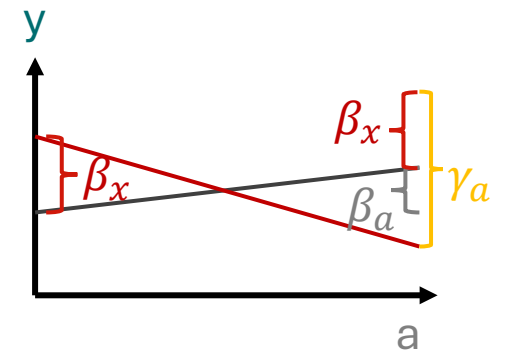
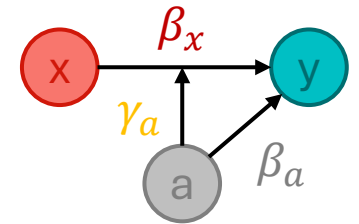
covariation, but
no moderation



no covariation,
but moderation



covariation and
moderation





Exercises

Applying Causal Modelling

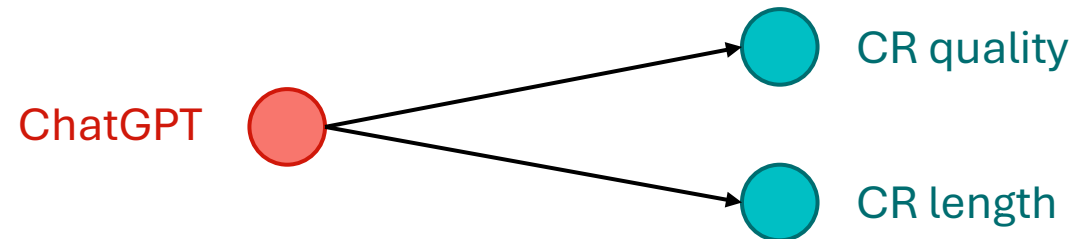
Exercise: Formulating a DAG

Derive a causal DAG from the following description:

“Researchers are interested in knowing if using ChatGPT affects the quality and length of code reviews at a company. Since they cannot interrupt the regular workflow, they cannot force anyone to use or not use ChatGPT. They do know, however, that younger developers use ChatGPT more often. On the other hand, they know that more skilled developers generally produce more concise (i.e., shorter) and better code reviews.”

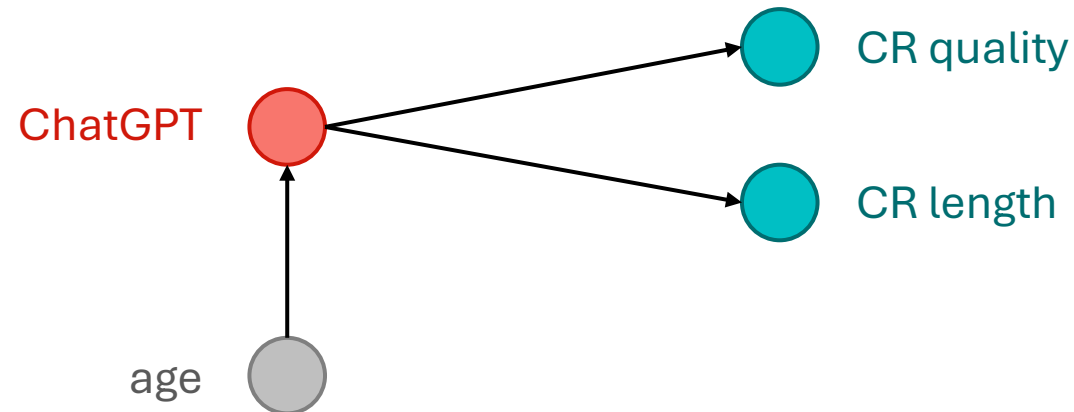
Solution: Formulating a DAG

“Researchers are interested in knowing if **using ChatGPT** affects the **quality** and **length of code reviews** at a company. Since they cannot interrupt the regular workflow, they cannot force anyone to use or not use ChatGPT. They do know, however, that younger developers use ChatGPT more often. On the other hand, they know that more skilled developers generally produce more concise (i.e., shorter) and better code reviews.”



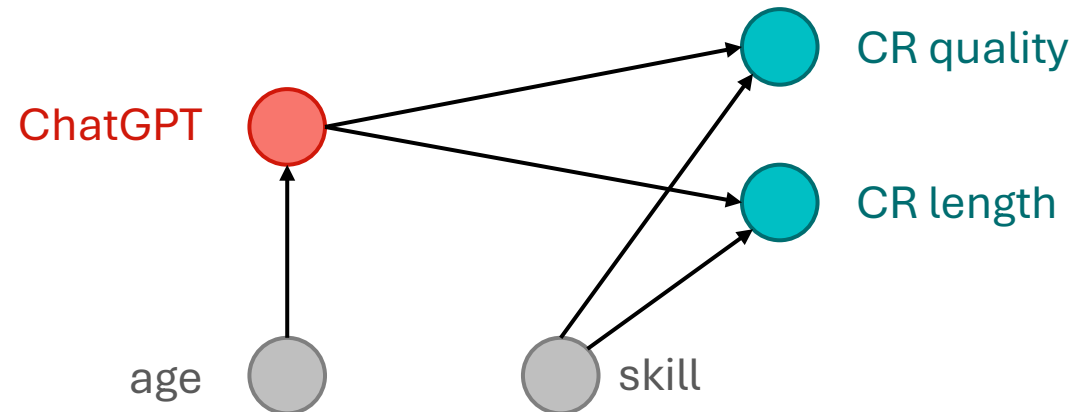
Solution: Formulating a DAG

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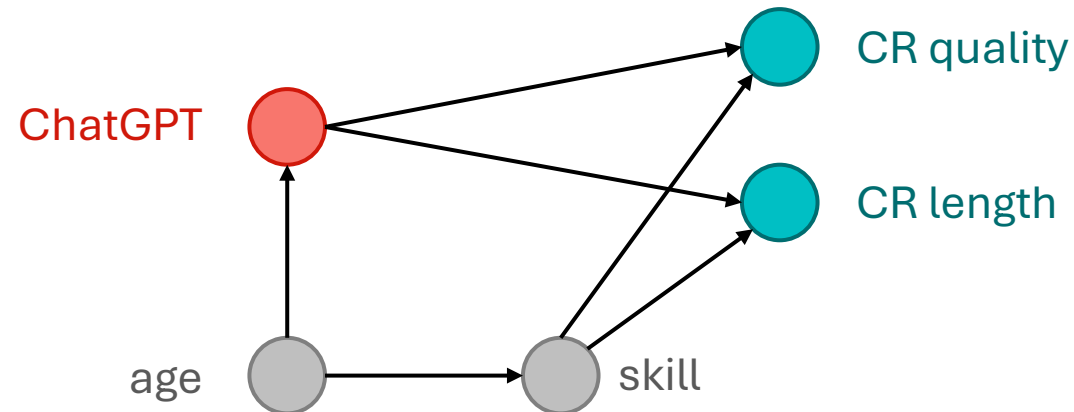
Solution: Formulating a DAG

“Researchers are interested in knowing if **using ChatGPT** affects the **quality** and **length of code reviews** at a company. Since they cannot interrupt the regular workflow, they cannot force anyone to use or not use ChatGPT. They do know, however, that **younger developers** use ChatGPT more often. On the other hand, they know that **more skilled developers** generally produce more concise (i.e., shorter) and better code reviews.”



Solution: Formulating a DAG

“Researchers are interested in knowing if **using ChatGPT** affects the **quality** and **length of code reviews** at a company. Since they cannot interrupt the regular workflow, they cannot force anyone to use or not use ChatGPT. They do know, however, that **younger developers** use ChatGPT more often. On the other hand, they know that **more skilled developers** generally produce more concise (i.e., shorter) and better code reviews.”



Material

All material is public and open-source on GitHub. Feel free to clone, fork, and share the study material.

Comments, requests, and additions are welcome.

JulianFrattini/bda4sci Public

The screenshot shows the GitHub repository page for JulianFrattini/bda4sci. The repository is public and has 4 stars, 1 watch, and 0 forks. The main branch is 'main'. The repository contains a file named 'sci-intro.Rproj' and a folder named 'version1'. The README file is visible, titled 'Introduction to Statistical Causal Inference', and mentions the Apache-2.0 license. The repository was last updated 5 months ago.

JulianFrattini / bda4sci

Code Issues 1 Pull requests Actions Projects Wiki Security Insights Settings

bda4sci Public Unpin Unwatch 1 Fork 0 Star 4

main 4 Branches 1 Tag Go to file Code

JulianFrattini Reorganized repository structure 9611265 · 5 months ago 23 Commits

File/Folder	Commit Message	Time Ago
version1	Reorganized repository structure	5 months ago
.gitignore	Reorganized repository structure	5 months ago
LICENSE	Initial commit	11 months ago
README.md	Reorganized repository structure	5 months ago
sci-intro.Rproj	Reorganized repository structure	5 months ago

README Apache-2.0 license

Introduction to Statistical Causal Inference

license Apache-2.0

This repository contains the material for the introduction seminar to Statistical Causal Inference (SCI). The purpose of the seminar is to introduce software engineering researchers with a background in analysis of quantitative data to a causal framework for inferential statistics proposed by Judea Pearl^[1] and Richard McElreath^[2]

About

Material for the Introduction Seminar to Bayesian Data Analysis for Statistical Causal Inference

bayesian-data-analysis
statistical-causal-inference

Readme
Apache-2.0 license
Activity
4 stars
1 watching
0 forks

Releases 1

Prototype Latest
on Oct 18, 2024

Packages

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References

- [1] Pearl, J. (2009). *Causality*. Cambridge university press.
- [2] Cinelli, C., Forney, A., & Pearl, J. (2024). A crash course in good and bad controls. *Sociological Methods & Research*, 53(3), 1071-1104.